25901388 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMAT204 - MATHEMATICAL TECHNIQUES IN BUSINESS ANALYTICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Explain the first order finite differences. [U] / [CO1]
2. Define Interpolation. [U] / [CO1]
3. What is meant by Inverse Interpolation? [U] / [CO2]
4. How forward interpolation differ from backword interpolation? [U] / [CO2]
5. Solve the quadratic equation $x^2 + 4x - 12 = 0$. [A] / [CO3]
6. Sketch the graph of P = -x² + 8x - 12, where P is profit and x is units sold. Find the vertex and intercepts. [A] / [CO3]
7. Find consumers' surplus if the demand function is $P = 120 - 4Q$ and the equilibrium values are $Q_0 = 6$and $P_0 = 96$ [U] / [CO4]
8. What is meant by constrained optimization? [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Following the pattern in the table, predict the value of y for the next step where x=40. [A] / [CO1]

$x:$	0	10	20	30
$y:$	15	30	50	75

10. Find the value of f(1.5) using newton's formula from this data: [A] / [CO1]

$x:$	1	2	3
$f(x):$	40	50	62

11. Calculate f(3) by applying Lagrange's formula to these points: [A] / [CO2]

$x:$	1	2	5
$f(x):$	1	4	25

12. The price-demand relationship of a product is given by $p = \frac{200}{x+5}$. [A] / [CO3]
    (a) Draw the graph of p versus x.
    (b) Identify the asymptotes and interpret them economically.

13. A business earns a monthly salary of ₹12,000 plus a commission of ₹25 per item sold. In one month, the total earnings were ₹17,000. [A] / [CO3]
    (a) Form a linear equation.
    (b) Find the number of items sold.

14. A company's total cost (C) in ₹ for producing x units of a product passes through the points (10, 5000) and (30, 11000). [A] / [CO3]
    (a) Find the equation of the cost line.
    (b) Determine the slope and y-intercept and interpret their economic meaning.

15. A firm produces two goods x and y. The individual cost functions are: $C(x) = 120 + 5x$ and $C(y) = 200 + 8y$. [E] / [CO4]
    (a) Form the total cost function $C(x, y)$.
    (b) Find the total cost when $x = 6$ and $y = 4$.

16. If marginal cost $MC = 2x^3 + 5x + 12$, find the additional total cost when output increases from 2 to 4 units. [K] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. If a graph represents a cubic polynomial $y = x^3$, how many columns of differences must you calculate before you find a constant value? Verify with: [A] / [CO1]

$x:$	0	1	2	3	4
$y:$	0	1	8	27	64

18. Evaluate f(3.2) by constructing a backward difference table for this data set. [A] / [CO1]

$x:$	1	2	3	4
$f(x):$	6	11	17	24

19. Find the quadratic polynomial P(x) for the data below and solve for x when P(x)=0: [A] / [CO2]

$x:$	0	1	2
$y:$	-3	-2	1

using polynomial interpolation.

20. Define the central difference operator δ and show how the table for the data below is constructed. Then, find the missing central value at x=0.5. [K] / [CO2]

$x:$	0	1	2	3
$y:$	1	2	4	7

21. A company's profit P (in ₹) from producing x units is given by $P = -2x^2 + 40x - 120$. [A] / [CO3]
    (a) Draw the graph of P versus x.
    (b) Find the vertex of the graph.
    (c) Find the x- and y-intercepts of the profit graph.

22. Evaluate the producers' surplus corresponding to the supply curve $P = 4Q^3 + Q^2 + 2Q + 6$,at the equilibrium values $Q_0 = 2$ and $P_0 = 50$ [E] / [CO4]

[10x3 = 30]